

Unified Lifelong Action Learning: A Real-Video Pipeline for Continual Learning, Knowledge Distillation, and Few-Shot Meta-Adaptation on UCF101

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Abstract—We present *Unified Lifelong Action Learning* (ULAL), an end-to-end pipeline that studies how a single video action recogniser can keep learning new actions over time without forgetting the old ones, trained directly on real video frames. The pipeline is built as five linked studies, each answering one question and each comparing several methods under one shared budget. (i) *How do we measure honestly?* UCF101’s shipped split leaks 80% of its source groups across train and test; we replace it with a group-disjoint split that leaks none. (ii) *Which backbone?* A five-way bake-off selects a ResNet50+LSTM teacher (93.9% on the base classes). (iii) *How do we stop forgetting?* On a $10 \rightarrow 15 \rightarrow 20$ class stream, rehearsal keeps 89.8% while naive fine-tuning collapses to 26.0% and the data-free methods (EWC, LwF) trail. (iv) *How do we shrink the model?* Cosine feature distillation gives a MobileNetV3-Small student that keeps 82.7% at $14\times$ fewer parameters. (v) *How do we learn new classes from a few clips?* A nine-arm comparison with a matched fine-tune control shows MAML+replay is the best-balanced few-shot adapter (harmonic mean 0.60). A final capstone chains the winner of every stage into one pipeline and tests it on real YouTube videos of five unseen sports: from only 3 clips per sport the MAML student reaches 0.30 accuracy on unseen videos against 0.04 for the plain distilled student, while with 5–8 clips the plain student catches up—meta-learning pays off exactly in the few-shot regime. The whole pipeline runs on a single free Colab T4 GPU.

Index Terms—continual learning, catastrophic forgetting, action recognition, knowledge distillation, meta-learning, few-shot learning, model compression.

I. INTRODUCTION

Deep video models forget. Fine-tuning a trained action recogniser on a set of new classes collapses its accuracy on the classes it already knew—*catastrophic forgetting* [1]. Lifelong, or continual, learning aims to add classes over time while preserving what was already learned [5]. Video makes this markedly harder than images: a clip is a large spatio-temporal tensor, so storing and replaying past data is expensive, and the temporal model on top of the per-frame features adds a second place for representations to drift.

A realistic deployment needs several abilities at once. It must (a) be measured on data it has never seen, (b) start from

strong features, (c) retain old classes as new ones arrive, (d) be small enough to run on a modest device, and (e) still pick up a brand-new class from only a handful of examples. Each of these is usually studied in isolation. We study them *together*, as one pipeline on one dataset, trained *directly on real video frames*, so every reported number reflects the model that would actually be deployed.

The pipeline (Fig. 1) is built on UCF101 [12] as five linked studies (S1–S5), each a notebook answering one question, and a capstone that chains the winner of every study. The organising idea is a *moving freeze boundary*: the recogniser is one shared network (a CNN applied to every frame, an LSTM over the frame features, and a classifier head that *grows* as classes arrive), and each stage decides which of its parts may still change.

Contributions.

- **An honest measurement protocol.** We quantify the 80% source-group leakage in UCF101’s shipped split and replace it with a group-disjoint split that leaks nothing.
- **A controlled study per stage:** five backbones, five continual-learning methods, four distillation objectives, and nine meta-adaptation arms, each drawn as its own schematic and compared under one shared budget and seed.
- **A fine-tune control that isolates meta-learning:** the MAML start is compared with an ordinary fine-tune start that saw the *same* extra data.
- **A real-video capstone:** the winners of all stages are chained into one pipeline and taught five unseen sports from a few YouTube clips, tested on YouTube videos they never saw.

II. RELATED WORK

Continual learning. Approaches fall into three families. *Regularisation* methods add a penalty that discourages changing parameters important to old tasks: EWC [1] weights the penalty by the Fisher information, and synaptic-intelligence

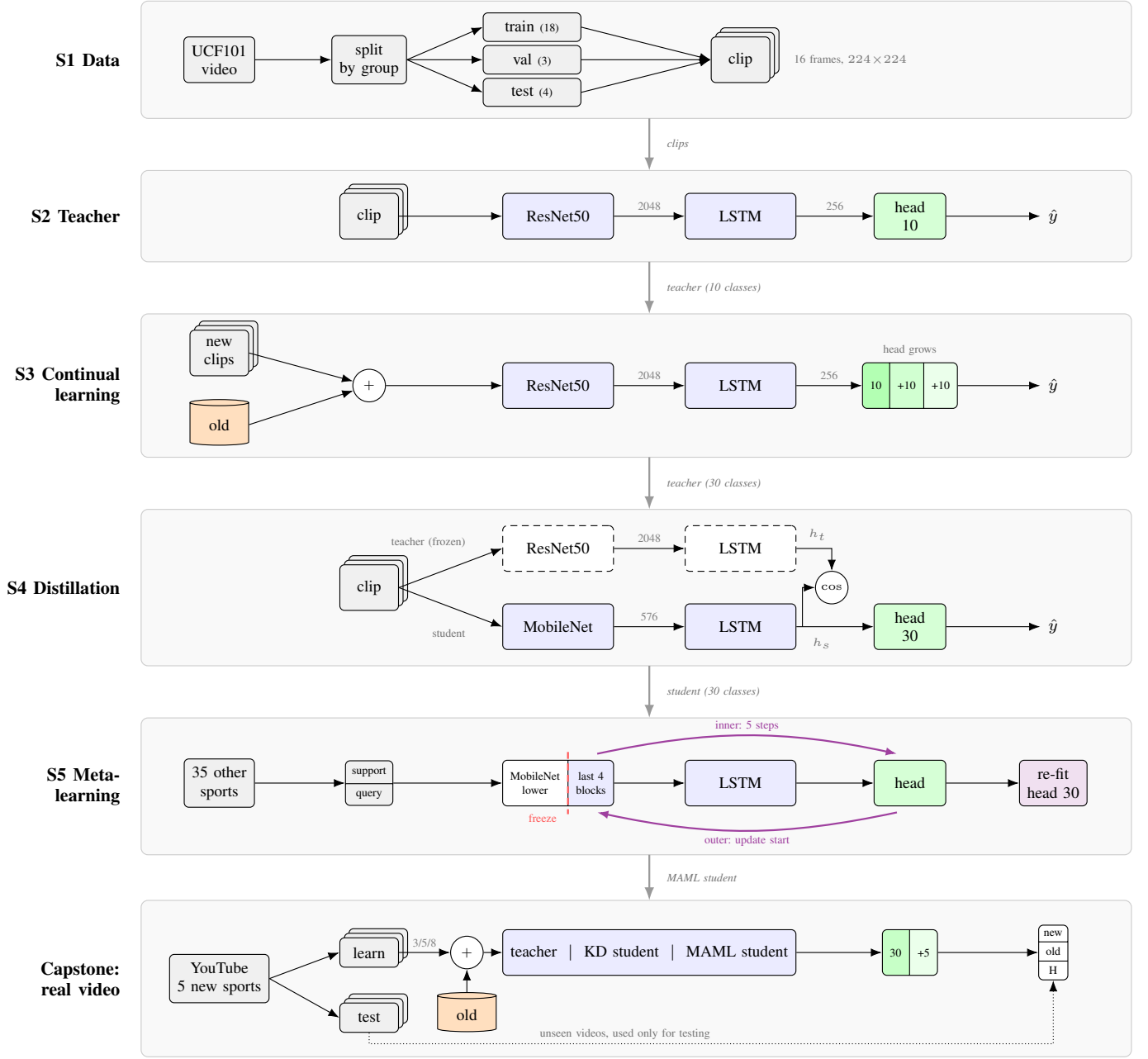


Fig. 1. **The ULAL architecture.** One panel per stage; the grey arrows show what each stage hands to the next. Every panel uses the winner of its study (Sections IV–VIII) and Section IX walks through the whole pipeline. Grey: data; blue: trained; dashed: frozen; orange: stored old clips; violet: meta-learning; green: classifier head; red dashed: freeze boundary.

variants use online importance estimates [2]. *Functional* methods keep the model’s outputs stable: Learning without Forgetting (LwF) [3] distills the old model’s own predictions on new data. *Rehearsal* methods store and replay a small set of past exemplars; iCaRL [4] combines rehearsal with distillation. We evaluate a representative of each family on video, where the cost of storing exemplars is a first-order concern. **Knowledge distillation.** A compact student can be trained to match a larger teacher’s softened logits [6] or its intermediate features [7]; we compare logit distillation against two feature objectives. **Meta**

and few-shot learning. MAML [8] learns an initialisation from which a few gradient steps adapt to a new task; first-order MAML drops the second-order term for efficiency. Few-shot class-incremental learning (FSCIL) [9] adds new classes from few examples while retaining a large base set, and reports the harmonic mean of base and new accuracy so that neither side can be ignored; we adopt that metric.

III. SYSTEM, NOTATION, AND PROTOCOL

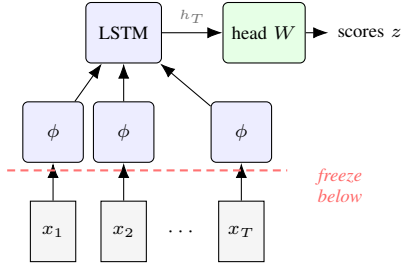


Fig. 2. **The shared recogniser.** The backbone ϕ runs on each of the 16 frames; an LSTM summarises them into a 256-d clip feature h_T ; a growing head classifies. The freeze boundary (dashed) marks how deep training may reach in a stage.

A. The shared recogniser

Every stage uses the same recogniser (Fig. 2). A clip is $T=16$ RGB frames at 224×224 , ImageNet-normalised. A CNN backbone ϕ is applied to each frame independently, producing frame features $\phi(x_1), \dots, \phi(x_T)$. An LSTM reads that sequence and its final hidden state $h_T \in \mathbb{R}^{256}$ is the clip feature. A linear head W maps h_T to class scores:

$$h_T = \text{LSTM}(\phi(x_1), \dots, \phi(x_T)), \quad z = W h_T. \quad (1)$$

The head *grows* as classes arrive: adding k classes appends k rows to W and keeps the existing rows. The teacher uses ResNet50 as ϕ (2048-d per frame), the student MobileNetV3-Small (576-d); both use the same 256-d LSTM, so their clip features h can be compared directly.

B. The moving freeze boundary

The stages differ in *which* parameters they train. Stage 2 trains the whole recogniser. Stage 3 trains the whole recogniser again for each new task (every method in that study trains the same parameters), and the head grows. Stage 4 freezes the teacher and trains the whole student. Stage 5 freezes the lower student backbone and meta-trains only its last four blocks, the LSTM, and the head, so the old features are protected while the upper layers reshape. Within a stage every compared method trains the *same* parameters, so only the mechanism under test differs.

C. Datasets and budget

All experiments use UCF101 on one free Colab T4 GPU. The group-disjoint split assigns whole source groups to one split, 18/3/4 groups per class (train/val/test), keeping the usual $\approx 72/12/16\%$ proportions (Section IV). Table I lists the budget of each stage; seeds are fixed and reset before every training run.

D. How to read the numbers

Test accuracy is the fraction correct on the unseen, group-disjoint test set. *Old* and *New* split it into classes learned earlier and the newest task; *Base* is retention on the original classes. When old classes outnumber new ones, a plain average mostly measures the old side and rewards a method that learns nothing

TABLE I
BUDGET PER STAGE (SHARED BY EVERY METHOD COMPARED IN THAT STAGE).

Stage	Key settings
Backbone (S2)	Adam, lr 10^{-4} , 5 epochs
Continual (S3)	Adam, lr 10^{-4} , 5 epochs/task; replay 15/class; LwF weight 3.0, $T=5$; EWC $\lambda=2800$
Distillation (S4)	Adam, lr 10^{-4} , 15 epochs; $T=2$; CE/transfer 0.75/0.25
Meta-train (S5)	5-way 3-shot, 60 episodes $\times$ 8 epochs; 40 sports; inner SGD 5 steps @ 0.05; meta-lr 10^{-4} ; last 4 blocks trained
Adaptation (S5)	8 clips/class, 5 epochs, lr 3×10^{-4} ; replay 8/class; LwF 0.5, $T=2$
Capstone	replay 15/class (10 $\rightarrow$ 20 $\rightarrow$ 30); cosine KD, 10 epochs; MAML 5-way 3-shot, 35 sports (1 clip/video); YouTube: 3/5/8 clips, 5 epochs, lr 3×10^{-4} , replay 8/class

TABLE II
SOURCE GROUPS SHARED BETWEEN SPLITS (20 STUDIED CLASSES, 2,679 VIDEOS).

Split	Groups	Groups in >1 split	Leak
Shipped (per clip)	500	400	80%
Group-disjoint	500	0	0%

new, so for few-shot learning we also report the harmonic mean

$$H = \frac{2 \text{acc}_{\text{new}} \text{acc}_{\text{old}}}{\text{acc}_{\text{new}} + \text{acc}_{\text{old}}}, \quad (2)$$

which is high only when *both* sides are high [9].

IV. STAGE 1 — DATA SPLIT AND LEAKAGE

Why this comes first. A UCF101 filename `v_Class_gXX_cYY.avi` encodes a *group* gXX (one source recording) and a *clip* cYY (one segment of it). All clips of a group share the same actor, background and camera. If the split is made per clip, segments of the *same* recording land in train and test, and the model can recognise the *scene* instead of the *action*. We therefore assign each whole group to a single split (panel S1 of Fig. 1).

Result and reading. Table II: the shipped split puts 400 of 500 groups in more than one split, so 80% of the test recordings were partly seen in training; the group-disjoint split puts none. It keeps the usual 72/12/16% proportions, so every later accuracy is lower than commonly reported on UCF101 only because the inflation is gone—and since the same leak would inflate every later stage, fixing it once at the source makes the whole study trustworthy.

V. STAGE 2 — BACKBONE SELECTION

Design. The backbone ϕ is the only part that changes: each candidate is plugged into the same recogniser (clip $\rightarrow \phi \rightarrow \text{LSTM} \rightarrow \text{head}$, panel S2 of Fig. 1) and trained with the same budget and seed. We expect a network with no prior knowledge

TABLE III

BACKBONE BAKE-OFF ON THE 10 BASE CLASSES (GROUP-DISJOINT TEST).

Backbone ϕ	Test acc. (%)	Params (M)	ms/clip
ScratchCNN	52.8	103.1	10.2
ResNet18	90.1	12.0	15.7
DenseNet121	92.0	8.3	53.1
VGG19-BN	92.0	20.8	115.7
ResNet50 (selected)	93.9	25.9	54.6

TABLE IV

CONTINUAL LEARNING AFTER THE FULL $10 \rightarrow 15 \rightarrow 20$ STREAM (%).
 FINAL: ALL 20 CLASSES; OLD: CLASSES 0–14; NEW: LAST TASK 15–19;
 BASE: ORIGINAL 10.

Method	Final	Old (0–14)	New (15–19)	Base (0–9)
S3.0 Naive	26.0	0.9	94.9	0.5
S3.2 LwF	30.5	5.3	100.0	0.0
S3.1 EWC	62.0	71.1	36.8	76.9
S3.4 Replay+LwF	87.0	82.6	99.2	83.0
S3.3 Replay	89.8	86.7	98.3	89.2

(a CNN trained from scratch) to struggle on a few thousand clips, and ImageNet-pretrained networks to cluster high.

Result and why. Table III. The from-scratch CNN has the most parameters (103M, because its LSTM reads a large flattened feature map) yet the worst accuracy (52.8%): capacity without useful priors does not help on little data. All four pre-trained backbones exceed 90%, because the hard part—good spatial features—is solved before the LSTM sees anything. **ResNet50 is selected:** it is the most accurate (93.9%), its 2048-d feature gives the later stages the richest representation, and at 55 ms per clip it is half as slow as VGG19-BN.

VI. STAGE 3 — CONTINUAL LEARNING

Design. Starting from the 10-class teacher, the head grows and the model learns two new tasks, $10 \rightarrow 15 \rightarrow 20$ classes. The five methods of Fig. 3 train the same network with the same budget; they differ only in *what protects the old classes*. Writing $\mathcal{L}_{CE}$ for cross-entropy on the batch:

- **S3.0 Naive:** $\mathcal{L} = \mathcal{L}_{CE}(\text{new})$ — nothing protects the past.
- **S3.1 EWC** [1]: adds $\frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2$, a penalty that keeps the weights θ important for old classes (Fisher F_i) near their old values θ^* .
- **S3.2 LwF** [3]: adds $\lambda \text{KD}(z_{1:C_{\text{old}}}, z_{1:C_{\text{old}}}^*)$, keeping the old-class scores of the new model close to those of a frozen copy of the old model, computed on the new clips.
- **S3.3 Replay:** stores 15 old clips per class and mixes them 1:1 into every new batch, $\mathcal{L} = \mathcal{L}_{CE}(\text{new} \cup \text{old})$.
- **S3.4 Replay+LwF** [4]: both.

Results. Table IV gives the final accuracies and Table V shows *when* each method forgets.

Why each method behaves as it does. *S3.0 Naive* learns every new task almost perfectly (98.2% on task 1, 94.9% on task 2) but the base falls from 93.9% to 22.6% after one task and to 0.5% after two: with only new labels in every batch, the shared weights are rewritten for the newest task. *S3.2 LwF*

TABLE V

WHEN EACH METHOD FORGETS: ACCURACY PER TASK (%) AFTER LEARNING TASK 1 AND AFTER LEARNING TASK 2. BEFORE ANY NEW TASK THE BASE ACCURACY IS 93.9.

Method	after task 1		after task 2		
	Base	Task 1	Base	Task 1	Task 2
S3.0 Naive	22.6	98.2	0.5	1.8	94.9
S3.1 EWC	80.7	79.1	76.9	60.0	36.8
S3.2 LwF	26.4	97.3	0.0	15.5	100.0
S3.3 Replay	89.6	94.6	89.2	81.8	98.3
S3.4 Replay+LwF	87.3	96.4	83.0	81.8	99.2

forgets almost as much (base 26.4% $\rightarrow$ 0.0%) even though it protects the old scores. The reason is the setting: the original LwF [3] gives every task its own head and is tested with the task known, whereas here one head covers all classes and the task is not known at test time. The distillation term only keeps the *ranking among old classes*, while the cross-entropy on new clips keeps raising the new classes above all old ones; the old clips are then out-voted by the newest classes—in the confusion matrices almost every forgotten base clip is predicted as a *new* class, not as another old one. *S3.1 EWC* fails the other way: its penalty keeps the base (76.9%) but also blocks the new task (36.8% on task 2), a stability/plasticity lock. *S3.3 Replay* is the only method that keeps every task high (base 89.2%, task 1 81.8%, task 2 98.3%): the stored clips come with their old labels, so every batch also tells the model what the old classes look like and puts them back in the competition. *S3.4 Replay+LwF* is close (87.0%) but slightly lower: the buffer already protects the old classes, and the extra distillation term only slows the new task. **Choice for the pipeline: replay** (panel S3 of Fig. 1). On video, storing a few clips beats every data-free method.

VII. STAGE 4 — KNOWLEDGE DISTILLATION

Design. The 25.9M-parameter teacher is too big for a small device, so we train a MobileNetV3-Small+LSTM student (≈ 1.8 M parameters, $14\times$ smaller). The four objectives of Fig. 4 share the student, data, budget and seed, and all use $\mathcal{L} = 0.75 \mathcal{L}_{CE} + 0.25 \mathcal{L}_{\text{transfer}}$:

- **S4.0 CE only:** $\mathcal{L}_{\text{transfer}} = 0$ (no teacher, the baseline).
- **S4.1 KD** [6]: match the teacher’s softened scores, $T^2 \text{KL}(\sigma(z_t/T) \parallel \sigma(z_s/T))$, $T = 2$.
- **S4.2 Cosine:** align the 256-d clip features by direction, $1 - \cos(h_s, h_t)$.
- **S4.3 MSE** [7]: a learnable regressor g maps h_s onto the teacher’s feature before an ℓ_2 match, $\|g(h_s) - h_t\|_2^2$ (g is dropped at inference).

Results and why. Table VI. The first lesson is *compression*: a student $14\times$ smaller stays within 4.3 points of its teacher (82.7% vs 87.0%), because the teacher’s decision function transfers. The second is the ranking of the objectives, which follows from what each one asks of the student. *S4.0 CE only* is already strong (80.6%): the student backbone is ImageNet-pretrained, so labels alone carry it far. *S4.1 KD* does not help

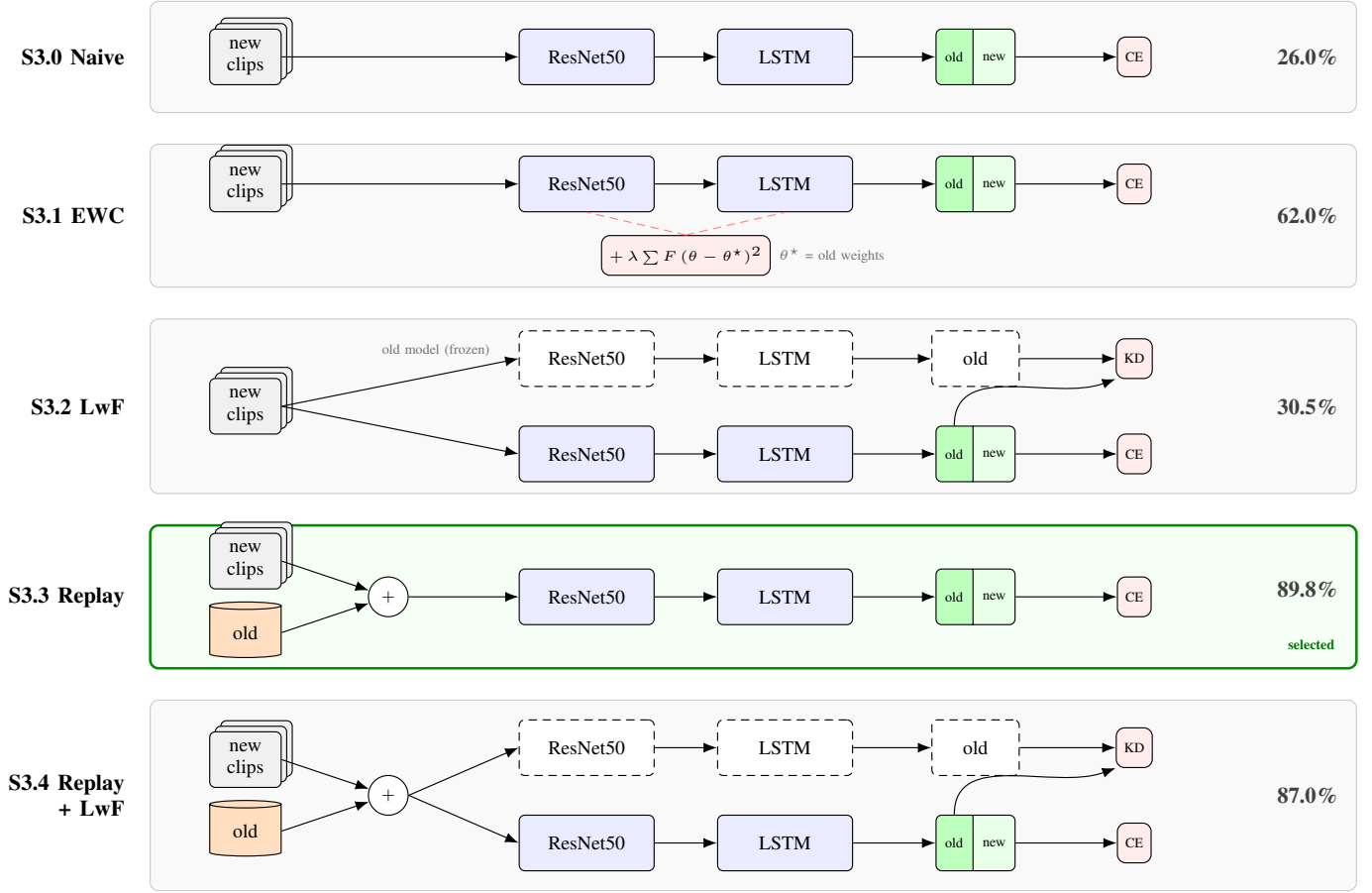


Fig. 3. **Stage 3: the five continual-learning methods.** All train the same network (blue) with a head that grows by the new classes; they differ only in what protects the old classes: nothing (S3.0), a weight penalty (S3.1), the scores of a frozen copy of the old model (S3.2, dashed), stored old clips (S3.3, orange), or both (S3.4). Right: final accuracy on all 20 classes (Table IV). Replay is used in the pipeline.

TABLE VI
DISTILLATION INTO THE MOBILENETV3-SMALL STUDENT (20-CLASS GROUP-DISJOINT TEST). SAME STUDENT, BUDGET, SEED AND INITIALISATION; ONLY THE TRANSFER TERM DIFFERS.

Model	Test acc. (%)	Params (M)
Teacher (ResNet50, Replay+LwF)	87.0	25.9
S4.0 Student, CE only	80.6	1.8
S4.1 Student, CE + KD	79.7	1.8
S4.2 Student, CE + cosine	82.7	1.8
S4.3 Student, CE + MSE (<i>g</i>)	75.9	1.8

here (79.7%): the softened teacher scores add little beyond the labels once the student is pretrained. *S4.2 Cosine* is best (82.7%, +2.1 over CE): it asks only that the student’s 256-d clip feature *point in the same direction* as the teacher’s, a gentle target that transfers the teacher’s temporal summary without forcing its scale. *S4.3 MSE* is worst (75.9%): it asks the student to match direction *and* magnitude across two very different backbones, a target that fights the label loss. The gap between the objectives is modest (single seed, 439 test clips), so the headline of this stage is compression, with cosine as the best transfer term. **Choice for the pipeline: cosine distillation**

(panel S4 of Fig. 1).

VIII. STAGE 5 — FEW-SHOT META-ADAPTATION

Design. Given the 20-class student, how well can it learn *five brand-new sports* from only a few clips? Every arm uses the same adaptation—the same 8 clips per new sport, 5 epochs, learning rate and seed, a head grown from 20 to 25 classes—so only two things vary (Fig. 5): the *starting point* and the *protection* of the old classes. The three starting points are:

- **S5.0 Plain:** the distilled student as it is.
- **S5.1 Fine-tune:** the student after ordinary supervised training on 40 other sports. This is the control: it sees the same extra data as MAML.
- **S5.2 MAML** [8]: the student after first-order MAML on the same 40 sports. Each episode draws 5 sports with 3 support and 3 query clips; the *inner* loop adapts a copy on the support clips, $\theta' = \theta - \alpha \nabla_{\theta} \mathcal{L}^{\text{sup}}(\theta)$ (5 SGD steps), and the *outer* loop moves the start by the loss of the adapted copy on the query clips, $\theta \leftarrow \theta - \beta \nabla_{\theta'} \mathcal{L}^{\text{qry}}(\theta')$.

For S5.1 and S5.2 only the last four backbone blocks, the LSTM and the head are trained, and a fresh 20-class head is then re-fitted to the new features (a short linear probe on 12

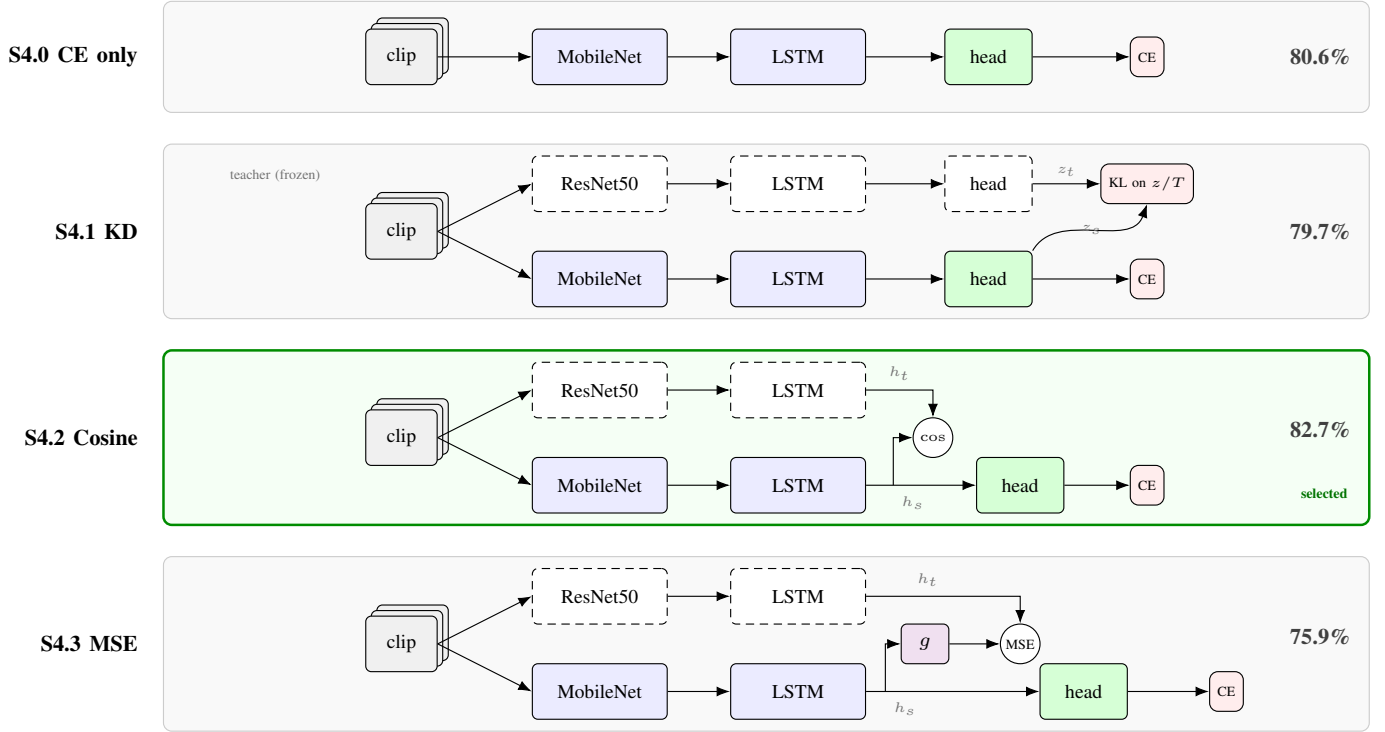


Fig. 4. **Stage 4: the four distillation objectives.** The student (blue) always learns from the labels (CE); the frozen teacher (dashed) adds nothing (S4.0), its softened scores (S4.1), its clip feature by direction (S4.2), or its clip feature through a learnable regressor g (S4.3). Right: student test accuracy (Table VI). Cosine is used in the pipeline.

TABLE VII

FEW-SHOT META-ADAPTATION: 20 BASE + 5 NEW SPORTS, 8 CLIPS PER NEW SPORT, IDENTICAL ADAPTATION FOR EVERY ARM, RANKED BY THE HARMONIC MEAN H (EQ. 2).

Start	Protection	New	Base	H
S5.2 MAML	replay	0.625	0.576	0.600
S5.2 MAML	replay+LwF	0.580	0.542	0.561
S5.1 Fine-tune	replay	0.464	0.617	0.530
S5.1 Fine-tune	replay+LwF	0.446	0.626	0.521
S5.0 Plain	replay	0.295	0.677	0.411
S5.0 Plain	replay+LwF	0.223	0.724	0.341
S5.0 Plain	none	0.527	0.189	0.278
S5.2 MAML	none	0.661	0.011	0.022
S5.1 Fine-tune	none	0.500	0.011	0.022

clips per class). Each start is adapted with no protection, with replay (8 old clips per class), or with replay+LwF, giving nine arms ranked by the harmonic mean of Eq. (2).

Results. Table VII ranks the nine arms; Table VIII shows how the new-sport accuracy grows with the number of clips. Before adaptation the student scores 75.9% on the base classes; during meta-training the query accuracy rises from 0.83 to 0.98 over 8 epochs, so the start is really being reshaped.

Why each arm behaves as it does. Without protection every start learns the new sports but wipes the base (base 0.011–0.189): with only new clips, the whole network is rewritten, exactly as in naive continual learning. *MAML alone* learns the new sports best (0.661), because its start is built

TABLE VIII

NEW-SPORT ACCURACY VERSUS THE NUMBER OF CLIPS PER NEW SPORT.

Start	Protection	1	3	5	10
S5.0 Plain	none	0.00	0.13	0.43	0.66
S5.1 Fine-tune	none	0.03	0.37	0.54	0.59
S5.2 MAML	none	0.26	0.42	0.55	0.65
S5.0 Plain	replay	0.00	0.04	0.23	0.45
S5.1 Fine-tune	replay	0.00	0.22	0.34	0.49
S5.2 MAML	replay	0.16	0.19	0.44	0.55

to move quickly onto a new task—and with nothing holding it back it moves all the way. *Replay* fixes the base in every case (0.576 to 0.724) but slows new learning, most for the plain start (0.295). *MAML+replay* combines a start that learns fast with stored clips that hold the old classes, and is the best balance (H 0.600). The key comparison is with the control: *fine-tune+replay* saw the same 40 sports and reaches only 0.530, so the extra 0.07 comes from meta-learning, not from the extra data. The LwF variants are always slightly worse: with only a few new clips, keeping the old scores fixed also holds back the fresh new-class rows. Table VIII shows that MAML’s advantage is a property of the *start*: it is largest with 1 clip (0.26 against 0.00–0.03) and disappears by 10 clips (0.65 against 0.66 for the plain start), when there are enough steps for any start to learn. **Choice for the pipeline: MAML meta-training, adapted with replay** (panel S5 of Fig. 1).

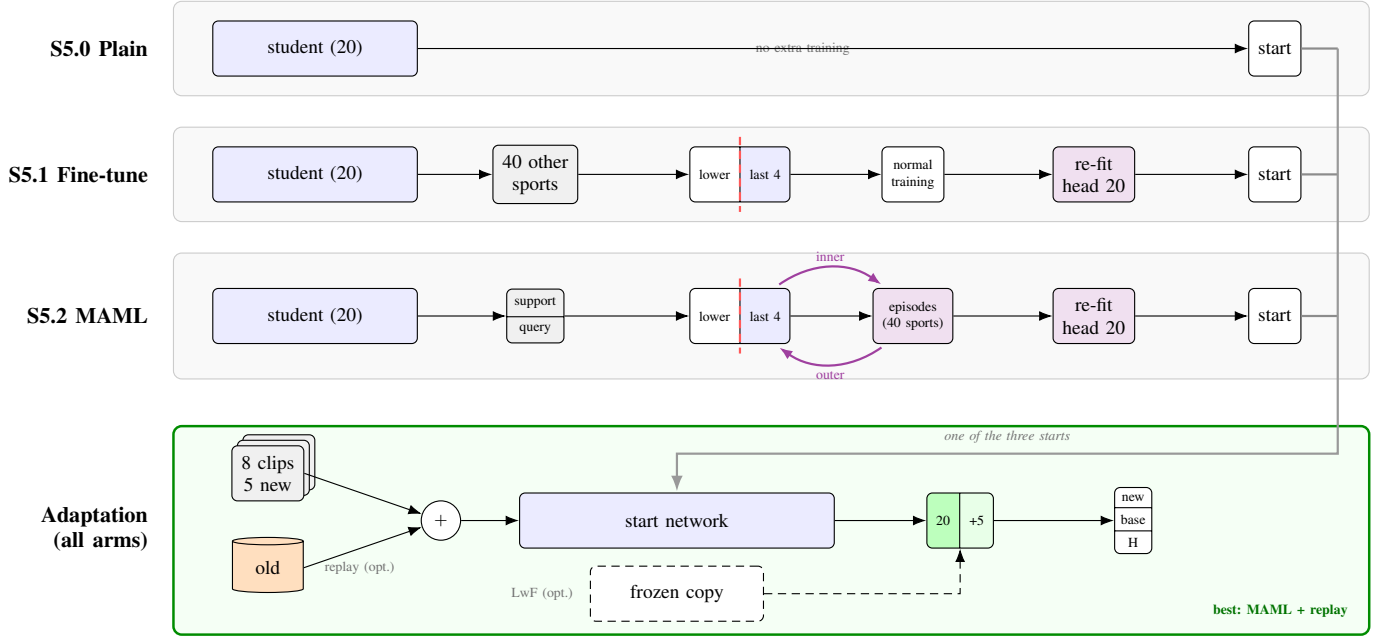


Fig. 5. **Stage 5: three starting points and one shared adaptation.** The plain student (S5.0), the student after normal training on 40 other sports (S5.1, the control), and the student after MAML on the same sports (S5.2, violet: inner/outer loop) are each adapted in the same way: 8 clips of 5 new sports, a head grown from 20 to 25 classes, with optional replay (orange) and optional LwF (dashed). Red dashed: only the last 4 blocks, LSTM and head are trained before adaptation.

IX. THE ULAL PIPELINE ON REAL VIDEO

Building the pipeline from the winners. The final notebook chains the selected method of every stage into one lifelong model, following Fig. 1 from top to bottom, now on a longer $10 \rightarrow 20 \rightarrow 30$ stream:

- 1) **S1 Data:** all clips come from the group-disjoint split.
- 2) **S2 Teacher:** the ResNet50+LSTM teacher, trained on the 10 base classes.
- 3) **S3 Replay:** the teacher’s head grows to 20 and then 30 classes; every batch mixes new clips with stored old clips (15 per class).
- 4) **S4 Cosine distillation:** the frozen 30-class teacher is distilled into the MobileNet student with CE + cosine.
- 5) **S5 MAML:** the student is meta-trained on 35 other sports—the 40 meta sports of Stage 5 minus the 5 that are among the 30 known classes—(5-way 3-shot, only its last 4 blocks, LSTM and head); each meta clip comes from a different source video, so the query clips are always from other videos than the support clips. Its 30-class head is then re-fitted on 12 clips per class.
- 6) **Real-video test:** 5 sports that no model has seen (golf swing, front crawl, hammer throw, handstand push-ups, handstand walking) are taken from YouTube, 4 videos each: 2 videos to learn from and 2 kept only for testing. Each model grows its head from 30 to 35 classes and learns from 3, 5 or 8 clips per sport with replay (8 old clips per class). We adapt the MAML student and, as references, the teacher and the plain distilled student, with the same clips, seed and budget. We report the accuracy on the unseen test videos (new), the accuracy

TABLE IX
THE PIPELINE ON UCF101 (GROUP-DISJOINT TEST).

Stage	Model	Classes	Acc. (%)	Params
S2	ResNet50 teacher	10	93.9	25.9M
S3	teacher + replay	30	79.7	25.9M
	old / new classes		78.8 / 81.4	
S4	student, cosine KD	30	67.2	1.8M
S5	MAML student (re-fitted)	30	59.9	1.8M

TABLE X
LEARNING 5 UNSEEN SPORTS FROM YOUTUBE CLIPS (WITH REPLAY).
NEW: ACCURACY ON THE UNSEEN YOUTUBE TEST VIDEOS; OLD:
ACCURACY KEPT ON THE 30 UCF CLASSES; H: HARMONIC MEAN.

Clips/sport	Model	New	Old	H
3	teacher	0.31	0.73	0.44
	KD student	0.04	0.62	0.08
	MAML student	0.30	0.53	0.38
5	teacher	0.28	0.79	0.41
	KD student	0.49	0.55	0.52
	MAML student	0.40	0.56	0.47
8	teacher	0.51	0.79	0.62
	KD student	0.62	0.62	0.62
	MAML student	0.42	0.53	0.47

kept on the 30 UCF classes (old), and their harmonic mean.

Results on UCF101. Table IX. On the longer stream replay still keeps the model balanced: after two tasks of ten classes

each the teacher scores 79.7% on all 30 classes, with old (78.8%) and new classes (81.4%) close together—no collapse towards the newest task. The distilled student keeps 67.2% at $14\times$ fewer parameters; the gap to the teacher is larger than in Stage 4 (30 classes instead of 20, and 10 instead of 15 epochs). MAML meta-training costs the student 7.3 points on the known classes (59.9%); moving the last blocks towards fast adaptation, and re-fitting the head on only 12 clips per class, trades some old-class accuracy for plasticity, as in Stage 5.

Results on real video and why. Table X. *With 3 clips per sport* the MAML student learns the new sports far better than the plain distilled student (0.30 against 0.04): this is exactly the regime MAML was trained for—a few clips and few update steps—and it matches the Stage-5 finding that the meta-learned start helps most when data is scarcest. *With 5 and 8 clips* the plain student catches up and passes it (0.49 and 0.62 against 0.40 and 0.42) and also keeps more of the old classes. Three reasons explain this: more clips mean more update steps, so the start matters less (as in Table VIII); the MAML student starts weaker on the known classes (59.9% against 67.2%); and adaptation here trains every layer with Adam for 5 epochs, further from the 5 small inner steps that MAML prepared for. *The teacher* keeps the old classes best (0.73–0.79) thanks to its larger capacity and 2048-d features, but learns the new sports slowly at the same small learning rate; at 8 clips it ties the plain student on H (0.62). Two limits of this test matter when reading it: there are only 10 test videos (2 per sport), so one video changes the new-sport accuracy by 0.10 and differences below that are within noise; and some sports change setting between videos—the handstand push-up learning videos are filmed in a white studio and the test videos in a gym, and at 8 clips the MAML student gets none of those test clips right. The clear, repeatable result is the 3-clip one: *meta-learning pays off when only a handful of clips exist*, while with more data the plain distilled student is the better deployable choice.

X. DISCUSSION AND LIMITATIONS

Three themes recur across the pipeline. *Measurement matters*: without the group-disjoint split every number would be inflated by scene memorisation, and without the harmonic mean the few-shot ranking would reward models that learn nothing new. *Data beats cleverness for retention*: on video, rehearsal of a few clips per class outperforms every data-free method (EWC, LwF); LwF in particular is designed for a setting where the task is known at test time and collapses when one head must choose among all classes. *Each tool has a regime*: distillation’s win is compression, and meta-learning’s win is learning from very few clips—with more clips the simpler student catches up, both in Stage 5 and on real video.

Limitations. All results use a single seed and one dataset, with short streams (up to 30 classes). Rehearsal stores real clips, so its memory cost grows with the number of classes. The real-video test is small (5 sports, 10 test videos) and depends on downloadable YouTube videos; a larger set of

videos and several seeds would make its differences at 5–8 clips conclusive.

XI. CONCLUSION

ULAL is a compact, honest, real-video pipeline for lifelong action recognition on a single free GPU. A leakage-free split makes the numbers trustworthy; a ResNet50+LSTM is a strong teacher (93.9%); rehearsal is the strongest continual learner (89.8% against 26.0% for naive fine-tuning); cosine distillation gives a $14\times$ smaller student that keeps 82.7%; and, scored with a harmonic metric and a fine-tune control, MAML+replay is the best-balanced few-shot adapter (H 0.60). Chained into one pipeline and tested on unseen YouTube videos, the meta-trained student learns new sports from 3 clips far better than the plain student (0.30 against 0.04), confirming that each chosen tool adds the ability it was selected for.

AUTHOR CONTRIBUTIONS

The work divides along the pipeline of Fig. 1, and Table XI indexes which author led which part. Each author owns a contiguous span of the pipeline together with a share of the shared infrastructure (the recogniser, the training harness, and the plotting and metrics code), and the two comparison-heavy stages (continual learning and meta-adaptation) sit with different authors. All three authors jointly designed the study, reviewed each other’s stages for fairness, and wrote the paper.

TABLE XI
AUTHOR CONTRIBUTION INDEX, ORGANISED BY PIPELINE STAGE.

Contributor	Responsibilities
Second B. Author	S1 data split and leakage audit; S2 backbone bake-off; the shared recogniser and training harness.
Third C. Author	S3 continual learning (all five methods, forgetting timelines); S4 knowledge distillation (four objectives, compression analysis).
First A. Author	S5 few-shot meta-adaptation (MAML, fine-tune control, harmonic metric); the ULAL capstone and real-video test; the evaluation protocol and metrics.

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